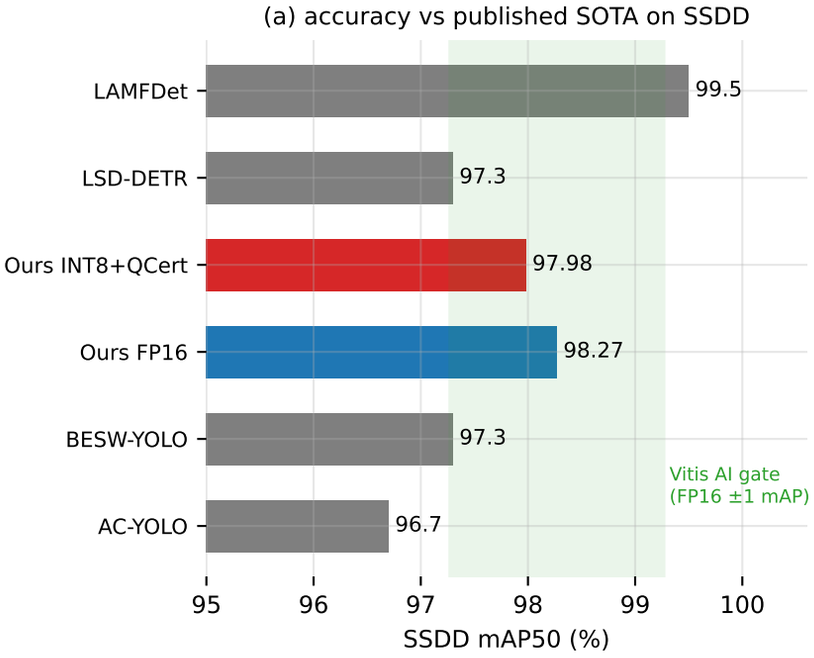
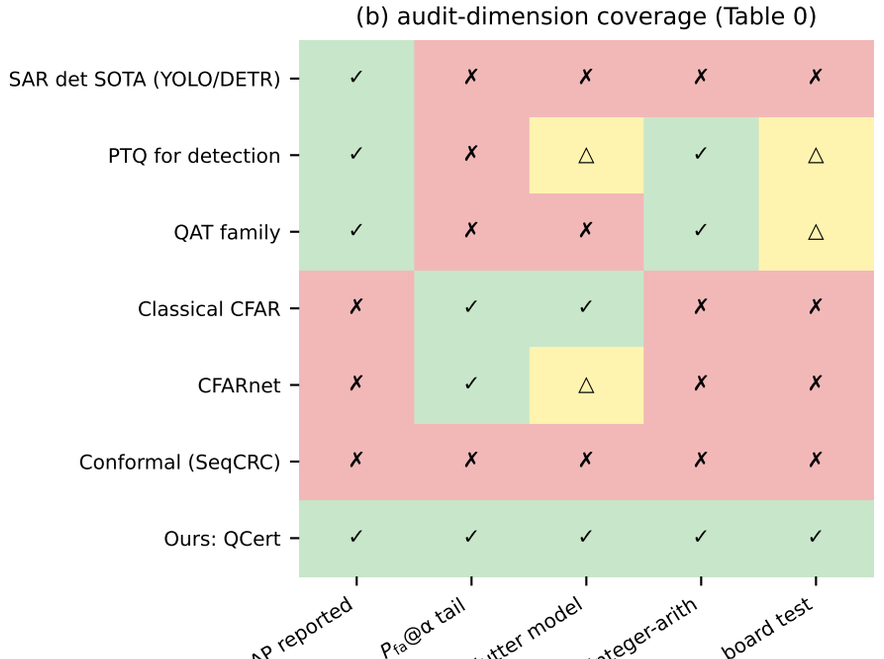
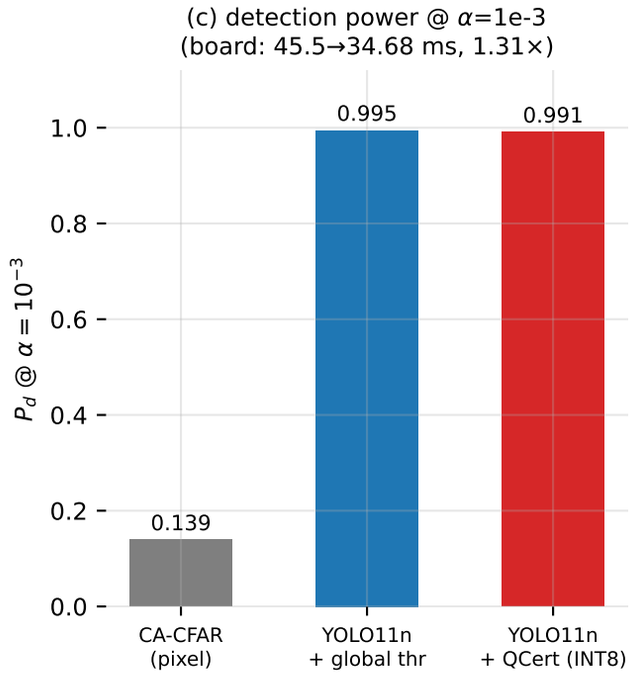

(a) accuracy vs published SOTA on SSDD
LAMFDet
LSD-DETR
Ours INT8+QCert
Ours FP16
BESW-YOLO
AC-YOLO
99.5
97.3
97.98
98.27
97.3
96.7
Vitis AI gate
(FP16 ±1 mAP)
95
96
97
98
99
100
SSDD mAP50 (%)


(b) audit-dimension coverage (Table 0)
SAR det SOTA (YOLO/DETR)
PTQ for detection
QAT family
Classical CFAR
CFARnet
Conformal (SeqCRC)
Ours: QCert
mAP reported
P_{fa}@α tail
clutter model
integer-arith
board test


(c) detection power @ α=1e-3
(board: 45.5→34.68 ms, 1.31×)
P_d @ $\alpha = 10^{-3}$
1.0
0.8
0.6
0.4
0.2
0.0
0.139
0.995
0.991
CA-CFAR
(pixel)
YOLO11n
+ global thr
YOLO11n
+ QCert (INT8)